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Energy and thermo-fluid-dynamics evaluations of photovoltaic panels cooled by water and air

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Abstract

The main limit of photovoltaic (PV) systems is the low conversion efficiency of cells, which is strongly influenced by their operating temperature. As the temperature increases the short circuit current (I_{sc}) increases moderately, while the open circuit voltage (V_{oc}) decreases considerably. The cell temperature reduction is a useful methodology that could be used in order to improve the PV panels performance of both new and already installed as well. This solution is interesting especially for high irradiation level locations with high external air temperature range along the daytime, because the maximum producibility occurs when the irradiation is high and therefore, as a consequence, the cell temperature increases. Furthermore, the proposed solution could be integrated with many PV typologies, already installed as well. Thus, it represents an alternative to PVT (Thermal–Photovoltaic) systems, which need DHW consumers for supplying the heat produced, otherwise the performance of the system will decrease. Various solutions adopting respectively a cooling water system and an airflow lapping the back of the panels in an open circuit is investigated to individuate the best cooling solution. Finite element software that describes with extreme detail the thermal exchange between the PV cells, the external environment and the cooling system is used in order to assess the reached temperature of the cells with different cooling system configurations calculating for each considered cases the overall thermal losses coefficient. Regarding the air cooling system configuration, which results less invasive, a comparison between the simulated and the measured, by laboratory tests, air speed has been conducted. Hourly energy simulations for the best configurations using the software TRNSYS are carried out to evaluate the annual performances.
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Keywords: Photovoltaic panel; Efficiency; Cooling system; Energy analysis

1. Introduction

Low efficiency of photovoltaic (PV) arrays is due to non-linear variations of output current and voltage as a function of solar radiation levels, operating temperature and load current. Hiyama et al. (1995) and Hua et al. (1998) described the interactions of those phenomena and overcame these problems using online or offline tracking algorithms in order to reach the maximum operating power point of PV systems forcing the system operating point toward the optimal condition. Furthermore, the main limit of PV system is the variation of the efficiency as a function of the operating. There are available in the literature various correlations showing both the electrical efficiency and the related power output of a PV module as a linear

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Nomenclature

A	cross section area, m^2
c	air speed, m/s
C_t	specific heat capacity, J/m^2 K
G_T	incident solar radiation, W/m^2
h_c	convective heat exchange coefficient between PV external surfaces and external air, W/m^2 K
I_{sc}	short circuit current A
k_{EVA}	thermal EVA conductivity, W/m K
k_g	thermal glass conductivity, W/m K
k_s	thermal Silicon conductivity, W/m K
k_{PET}	thermal PET conductivity, W/m K
k_{ted}	thermal Tedlar conductivity, W/m K
NOCT	nominal operating cell temperature, °C
s_{EVA}	thermal EVA conductivity, W/m K
s_g	glass thickness, m
s_s	silicon thickness, m
s_{PET}	PET thickness, m
s_{ted}	Tedlar thickness, m
Q_{el}	specific electricity production, W/m^2
Q_{loss}	specific thermal power losses, W/m^2
Qr_{ad}	specific collected solar power, W/m^2
T_{env}	external air temperature, K
U_{pan}	overall heat loss coefficient, W/m^2 K
$U_{pan\text{-}system}$	overall heat loss coefficient of PV-cooling system, W/m^2 K
V_{mp}	tension at maximum power, V
V_{oc}	open circuit tension, V
α	solar absorption coefficient, -
η	conversion efficiency, %
λ	thermal conductivity, W/m K
ρ	density, kg/m^3
τ	solar transmission coefficient, -

function of the cell operating temperature for PV modules or PV arrays mounted on free-standing frames, for PV/thermal collectors and for BIPV arrays, respectively (Skoplaki and Palyvos, 2009). As Fig. 1 shows, the output power of a single-crystalline silicon solar cell decreases as its temperature increases. Usually the temperature coefficient of the maximum output power for crystalline silicon PV operating above 25 °C is −0.65%/K (Raziemska, 2003), in accordance to the efficiency reference value reported in literature that is approximately 0.06 in absolute value per kelvin degree increase (Joshi, 1994). Malik (2003) analyzed the outdoor performance of a PV system; the results show that cold temperatures produce more efficient photo conversion for monocrystalline solar cells. Therefore, in tropical regions keeping as low as possible the PV module temperature is paramount to optimize PV performance (Ye et al., 2013).

Studies showing the benefits of PV modules cooled by both air and water have been carried out by many scientists. As regard air as working fluid, many studies were done (Bambrook and Sproul, 2012; Brinkworth et al., 1997), especially concerning building integration purposes (Yun et al., 2007; Ghani et al., 2012a, 2012b; Yin et al., 2013). Instead regarding water as working fluid, studies of hybrid concepts were done (Assoa et al., 2007; Anderson et al., 2009; Mishra and Tiwari, 2013) but focused on enhancing PV performance as well, indeed Amery and Abdolizadeh (2009) investigated the effect of spraying water upon the PV cells in terms of cells temperature variation and reflection losses showing an increase in cells efficiency on average of 3.26% and an increase in optical performance of 1.8% compared with standard operating mode. (Krauter, 2004) examined a cooling strategy consisting of a flowing film of water on the PV module front; the results showed an improvement of the optical performance by 1.5% and an increment of the electrical energy yield over the whole considered day by 10.3% compared to the same PV system without cooling. In conclusion the previous studies show the convective effect of air and water improves the working condition of the PV system as a consequence of decreasing cells temperature. Based on this results the aim of the investigation is to study the effects of a simple air and water cooling systems on the electrical efficiency of a PV module by a duct located on the back of PV panel without exceeding the frame dimension and thus allowing the installation even in the existing one. An energy comparison of the best presented PV panel – cooling system solutions are shown afterwards.

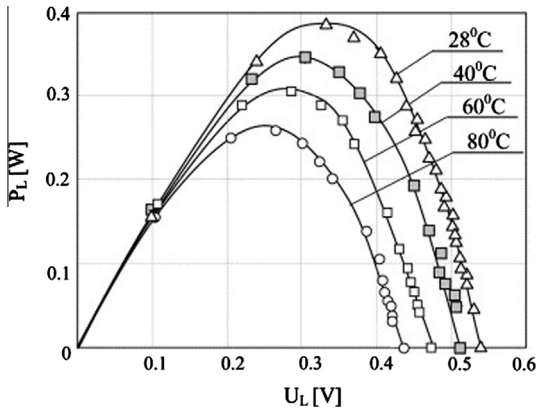


Fig. 1. Output power versus voltage of a single-crystalline silicon solar cell at various temperatures (Raziemska, 2003).

2. Methodology

The PV panel used in this research consists of the following layers from the top to the bottom:

Tempered prismatic glass with low Fe content with anti-reflective coating, Ethylene – Vinyl – Acetate EVA

encapsulated (transparent), 36 polycrystalline silicon cells, high efficiency square typology, size 156 mm × 156 mm, transparent Tedlar anti-moisture film, Polyethylene Terephthalate, Aluminum. All layers are looked in an anodized aluminum frame. Thermal properties regarding the above materials are shown in the Table 1.

The solar radiation transmission coefficient τ of the glass is 0.9 and the silicon has 0.85 as solar absorption coefficient α. The size of the considered PV panel is 1041 × 991 × 35 mm and its weight is 14.5 kg. The electrical features of the PV panel related to the standard conditions (STC) are as follow: maximum power 140 W, open circuit voltage 22.35 V, voltage at maximum power 17.84 V, short circuit current 8.20 A, current at maximum power 7.84 A, filling factor 76.40%, efficiency 13.60%. The working performances, as a function of cell temperature, are expressed by the related coefficients: +0.05%/°C regarding the short circuit current, −0.34%/°C regarding the open circuit voltage and −0.43%/°C regarding the power. The overall heat losses coefficient (U_{pan}) gives information about the heat removed from the PV cells. This parameter is not directly provided by the producer but it is connected to the nominal operative temperature of cell (NOCT) which is the temperature of the cells at the following working conditions (Cucumo et al., 1994): 800 W/m^2 solar incident flux, external air temperature of 20 °C, wind speed of 1 m/s, module working at open circuit condition. The NOCT of the cell provided by the manufacturer is 47 °C. Usually NOCT conditions are used to compare different PV systems; therefore the analysis of combined PV-cooling systems has been referred to them using for the heat exchange between the external surfaces and the environment the convective coefficient h_c related to NOCT conditions. In particular, these conditions are used to estimate the global heat loss coefficient U_{pan} thought the formula below:

$$(\tau\alpha)\cdot G_T = U_{pan}\cdot(NOCT - T_{env}) \quad (1)$$

Firstly the conventional PV panel by means of the finite elements software has been analyzed. Using the materials shown in Table 1 and imposing the boundary conditions related to the aforementioned NOCT condition, the convective heat exchange coefficient between PV external surfaces and external air has been estimated verifying (1). Hence, stating U_{pan} as:

$$U_{pan} = \frac{1}{\frac{1}{h_c} + \frac{s_g}{k_g} + \frac{s_{ev}}{k_{ev}} + \frac{s_s}{k_s} + \frac{s_{EVA}}{k_{EVA}} + \frac{s_{ted}}{k_{ted}} + \frac{s_{PET}}{k_{PET}} + \frac{s_{ted}}{k_{ted}} + \frac{1}{h_c}} \quad (2)$$

The coefficient h_c has been iteratively evaluated matching the NOCT value. Instead concerning the modeling of the cooling system located on the back of the PV panel for all the considered solutions, the inlet fluid temperature and the convective heat transfer coefficient between the cooling ducts surfaces and the fluid have been fixed accordingly to the literature value.

In regards to the water cooling system a close loop with a water–air heat exchanger has been supposed, therefore the inlet temperature of the water has been set, in the finite elements model, higher than the external air temperature of 10 K. The convective heat flux between the cooling ducts and the working fluid has been estimated considering a water speed of 0.2 m/s and 1.5 m/s corresponding to laminar and turbulent flow as shown in Table 2, which shows the convective coefficients and Reynolds numbers related to each flow condition (Marinelli et al., 1984). The correlations used to evaluate the convective coefficients have been implemented in the model assuming the fluid speed orthogonal at the cross section of each cooling duct.

Instead, in regards to the air cooling system, an adiabatic cooling at the inlet of the relative duct has been supposed; therefore the inlet air temperature has been fixed, in the finite elements model, equal to the external air temperature value. Concerning the convective heat transfer coefficient within the cooling duct, a comparison between measured and simulated velocity field of the working fluid has been taken into consideration. A mean speed of 2.3 m/s orthogonal at the cross section of each cooling duct has been assumed in the simulation; as a consequence the Reynolds number assumes the value of 6199 and the associated convective heat transfer coefficient of 21 W/m^2 K (Cucumo et al., 1994; Marinelli et al., 1984).

In the finite element models the working fluids properties, thermal conductivity, specific heat, viscosity and density has been set accordingly to the literature values (Incropera and DeWitt, 1996; Bejan, 1993).

Therefore, concerning all the considered PV-cooling system solutions, the overall heat losses coefficient $U_{pan\text{-}system}$ has been calculated using the cell temperature value achieved by means of the finite element software for each configuration. Moreover concerning the external surfaces

Table 1
PV module stratigraphy and thermo-physical properties of the materials.

Material	Density (kg/m^3)	Thermal conductivity (W/m K)	Specific heat (J/kg K)	Thickness (cm)
Glass	2500	1.16	840	0.4
EVA	960	0.25	2090	0.018
Silicon	2336	149.00	700	0.027
EVA	960	0.25	2090	0.018
Tedlar	1780	0.20	1050	0.011
PET	1330	0.20	1047	0.015
Tedlar	1780	0.20	1050	0.011
Aluminum	2700	160.00	900	–

Table 2
Water cooling system. Flow characteristics and convective heat transfer coefficients.

	Speed (m/s)	Reynolds	h_{conv} (W/m^2 K)
Laminar	0.2	1600	365
Turbulent	1.5	12,000	7351

of the whole system, NOCT conditions and the external convective coefficient h_c have been set as boundary conditions. Thus, to asses $U_{pan\text{-}system}$ the NOCT Eq. (1) has been used replacing the NOCT with the achieved cell temperature:

$$U_{pan-system} = \frac{(\tau \cdot \alpha) \cdot G_T}{(T_{cell} - T_{env})} \quad (3)$$

Once evaluated the thermal properties of the proposed PV panel – cooling system configurations, the PV energy surpluses due to the different cooling systems through the energy simulation code TRNSYS® have been estimated (TRNSYS, 2006). Furthermore the type 180, appropriate to describe a PV-cooling system, has been used to carry out the energy analysis. The modeling of the system performance could occur by using three different options (Mathematical Reference, 2006):

a. Assuming the cell temperature through an external file.
b. Providing the NOCT value utilized in (1) to estimate the loss coefficient.
c. Providing directly the overall heat loss coefficient U_{pan}.

In order to meet the previous analysis and to consider the effect of the thermal capacity the third option has been chosen (Mathematical Reference, 2006). Hence, the equations used to model the PV-cooling system in TRNSYS are:

$$C_t \cdot \frac{dT}{dt} = Q_{rad} - Q_{el} - Q_{loss} \quad (4)$$

$$Q_{rad} = (\tau\alpha) \cdot G_T \quad (5)$$

$$Q_{el} = \eta \cdot G_T \quad (6)$$

$$Q_{loss} = U_{pan} \cdot (T_{cell} - T_{env}) \quad (7)$$

where C_t is the thermal capacity, G_T is the incident solar power on the PV panel, Q_{rad} is the fraction of the solar power absorbed by the silicon layer, Q_{el} is the electrical power produced by the PV panel and Q_{loss} is the lost thermal power. In order to compare the various cooling systems four representative days have been considered: a summer and a winter day with clear sky conditions, a summer and a winter cloudy day. The referring climate data for the energy simulation are related to the city of Cosenza (Latitude 39.3°N, Longitude 16.25°E) representatives of Mediterranean climate. Moreover yearly simulations, based on hourly time-step, have been conducted for the best refrigeration system configurations estimating the PV energy production affected by the cooling effect. The energy consumption of fluid circulation systems for both air and water have been not considered in this analysis.

3. Experimental procedure

3.1. Water cooling system

Referring to Fig. 2, which shows a comprehensive picture of the cooling system, the system consists of a series of square ducts on the back of the PV panel, in which the working fluid flows. In order to increase the contact exchange surface between the ducts and the PV panel a square section has been chosen. In the preliminary phase a pitch between the cooling ducts has been fixed to 52 mm (Fig. 2) afterwards a sort of optimization study of the best pitch configuration has been carried out.

Polypropylene and aluminum as cooling ducts materials, and an aluminum sheet between the PV panel and the cooling system have been tested.

The purpose of the metal sheet is to uniformly cool the back of the PV panel. Polypropylene is not a good thermal conductor, $\lambda = 0.22$ W/m K, but its low weight allows a simple mounting on the back of the PV panel and it is cheaper than the aluminum. Nine proposed solutions are listed in Table 3: with or without metal sheet with different thickness. The first listed case is the conventional one without any cooling system, taken as a reference. Laminar and turbulent regimes have been considered as well. In particular the cases regarding the turbulent regime are indicated with the superscript in the results section.

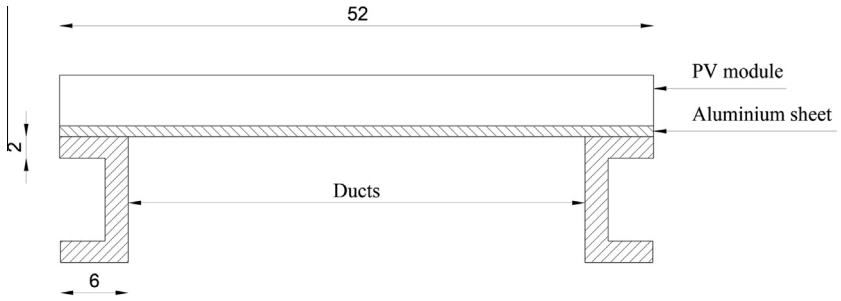


Fig. 2. Cross section of a portion of the PV-cooling water system (measure in mm).

Table 3
Different considered configurations of the PV water cooling system.

	Duct material	Sheet thickness (mm)
Case 1	–	–
Case 2	Polypropylene	–
Case 3	Aluminum	–
Case 4	Polypropylene	1
Case 5	Aluminum	1
Case 6	Polypropylene	0.5
Case 7	Polypropylene	2
Case 8	Aluminum	0.5
Case 9	Aluminum	2

3.2. Air cooling system

The air system showed in Fig. 3 consists of aluminum ducts on the back of the panel covering the entire back surface of the PV panel. The inlet and outlet openings are at the opposite sides along the longer length of the duct, thus the air flow affects the largest possible surface of the PV panel as shown in Fig. 4, which shows a simple drawing of one cooling duct. The inlet and outlet openings have a square section with around 120 mm as side length (Fig. 4).

A cooling duct section that maximizes the air flow without exceeding the PV frame has been chosen accordingly to the literature (Brinkworth, 2006). The analysis has considered different number of cooling ducts, changing their size in order to maintain the back surface of the PV panel affected by the airflow.

Four system configurations have been taken in account:

- Case A1 – Three ducts with air directly in contact with Tedlar layer.
- Case A2 – Three ducts with 1 mm aluminum sheet in contact with Tedlar layer.
- Case A3 – Three ducts with 2 mm aluminum sheet in contact with Tedlar layer.
- Case A4 – Four ducts with 1 mm aluminum sheet in contact with Tedlar layer.

3.3. Air speed field validation

Concerning the air cooling system, the assumed mean speed of 2.3 m/s has been verified by means of a finite elements analysis following a similar approach used by Brinkworth et al., (1997). A cooling duct model made in wood according with the dimensions of the A4 case has been created (Fig. 4). The interior surface of the duct is lacquered in order to produce the same smooth surface of the aluminum. A series of circle holes, with 11 mm of diameter, on the longitudinal cross section of the duct have been made as shown in Fig. 4. The holes position along the three sections is asymmetric in order to achieve for each section the information on five measure points.

The hot wire anemometer BSV 102 of LSI® Company has been used to get the speed measures. It has an accuracy of 3% when the speed is included in the range 0–30 m/s at 25 °C (Probes Air speed Manual, 2005). A simple 120 × 120 × 25 mm helical fan of 3.6 W Power located on the outlet section has been utilized to blow off the air from the cooling duct. The experimental model has been reproduced by means of a fluid dynamic software, assigning to the outlet square section a 1.15 m/s speed. That speed has been estimated by mass balance considerations as below:

$$\rho A_s c_s = \rho A_{out} c_{out} \tag{8}$$

where A_s is the cross section area of the duct, c_s is the cross section mean speed, A_{out} is the outlet section area of the

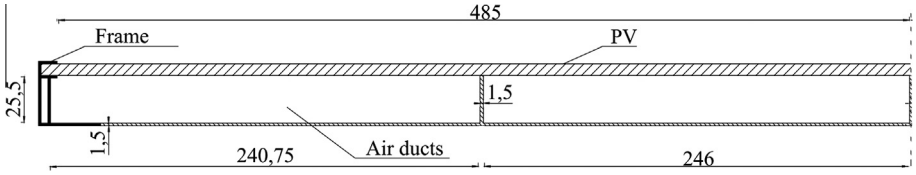


Fig. 3. Cross section of an half of the PV-cooling air system (measure in mm).

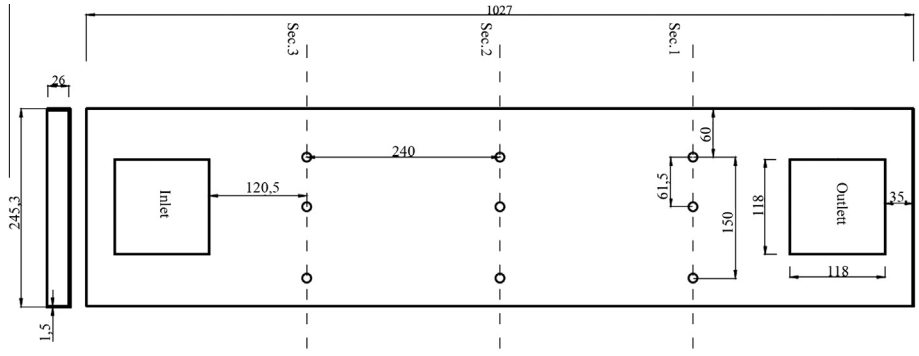


Fig. 4. Simple drawing of one air experimental model cooling duct including measure points (measure in mm).

duct, c_{out} is the outlet section mean speed and ρ is the air density.

4. Results

4.1. Cooling performance of water systems

Table 4 shows the PV mean cell temperature values, the overall heat losses coefficient and the thermal capacity of the whole coupled system, PV panel – cooling system, achieved with the configurations illustrated in Table 3 using the finite element software.

The thermal performance of the cooling system is extremely influenced by the pipe material (Cases 2 and 3). Concerning the cases with metal ducts, the coefficient $U_{pan\text{-}system}$ is always greater than 40 W/m^2 K. The turbulent flow condition compared with laminar flow condition brings considerable beneficial effects especially for the cases with metal ducts. The effect of the aluminum sheet is positively noticeable only with metal pipes (Cases 5–8–9, vs. Case 3), in which cases the combined effect of duct and sheet produces a lower temperature difference between the ducts as showed in Fig. 5, which shows a comparison between Cases 3 and 5. Moreover the heat rejection effect of the PV panel frame is boosted by the metal sheet, which transfers the heat towards the aluminum frame exploiting its high thermal diffusivity (Fig. 5).

Increasing the metal sheet thickness weakly improves the performance of the system, even turbulent flow regime (Table 4); therefore sheet thickness greater than two millimetres has not been considered in order to avoid gaining weight by the PV-cooling system. The metal sheet promotes the heat exchange decreasing the temperature difference along the back of the PV panel.

Fig. 6 shows the temperature profile of some PV-water cooling system configurations concerning the optimization phase. In particular, it regards the following configurations: two ducts located at the PV frame edges, two ducts located at the PV frame edges and one on the center, five ducts placed equidistant from the center. The temperature difference between the edge and the middle of the panel is more than 15 K for the first configuration; moreover at the middle of the panel (0.50 m distance from the edge) the temperature is close to the NOCT value, this configuration shows only effects of cooling along the edges of the panel (Fig. 6). Using an additional cooling duct located on the center of the PV panel, the PV cells temperature decreases and the maximum value results lower than the NOCT temperature (Fig. 6).

The aim of the optimization of the pitch is to achieve an almost flat PV cells temperature profile using few cooling ducts possible.

The configuration, result of the optimization process, consists of five cooling ducts of which: one is located at the center of the PV panel and the others are situated from this with a pitch of 159 mm. Fig. 7 shows the cross section of an half of the PV-water cooling system final configuration. The cell temperature profile of the optimized configuration has a maximum variation of 3 K (Fig. 6), the

Table 4
Mean cell temperature (T_{cell}) and overall heat loss coefficient (U_{pan}) of the different configurations of water cooled PV panels. The results concerning the turbulent regime are indicated with the superscript.

	T_{cell} (K)	$U_{pan\text{-}system}$ (W/(m^2 K))	C_t (J/(m^2 K))
Case 1	320.35	22.52	10,184
Case 2	314.55	28.66	10,184
Case 3	308.25	40.52	10,184
Case 4	313.95	29.43	12,614
Case 5	305.95	47.93	12,614
Case 6	313.95	29.38	11,399
Case 7	313.95	29.46	15,044
Case 8	306.05	47.39	11,399
Case 9	305.85	48.25	15,044
Case 2′	313.85	29.56	10,184
Case 3′	307.55	42.36	10,184
Case 4′	313.15	30.59	12,614
Case 5′	304.75	52.71	12,614
Case 6′	313.15	30.52	11,399
Case 7′	313.15	30.63	15,044
Case 8′	304.95	51.93	11,399
Case 9′	304.65	53.17	15,044

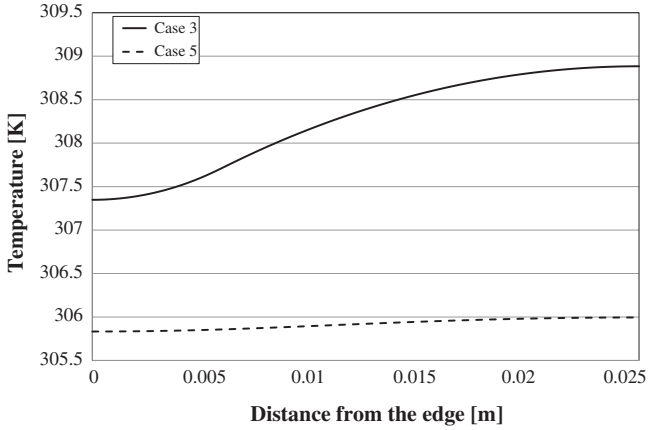


Fig. 5. PV cells temperature profile of PV-water cooling system with aluminum ducts with (Case 5) and without (Case 3) metal sheet.

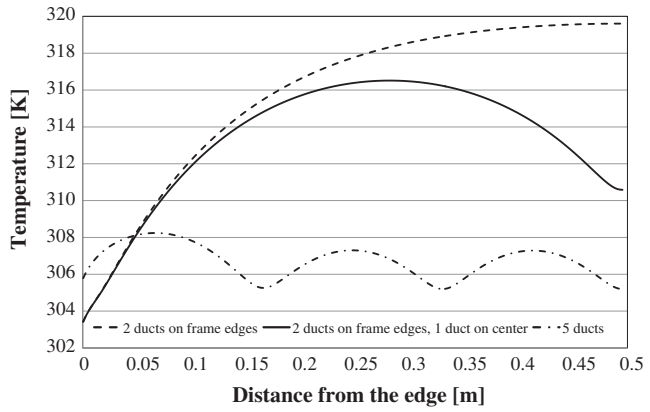


Fig. 6. PV cells temperature profile of different PV-water cooling system solutions regarding the optimization phase.

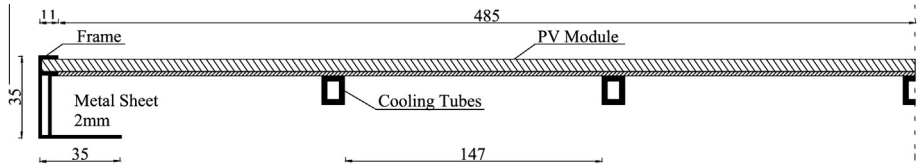


Fig. 7. Cross section of half PV-water cooling system of the optimized solution (measure in mm).

temperature is moderately higher near the frame because of the absence of cooling ducts in this zone. The overall heat losses coefficient of the whole PV panel – cooling system configurations of the optimization process as a function of ducts number and ducts location is shown in Fig. 8. The ducts in each solution are equidistant from the center of the PV panel.

4.2. Cooling performance of air system

Table 5 shows the results obtained by using the finite element software; in particular, the mean PV cell temperature values, the overall heat losses coefficients and the thermal capacity of the PV panel – cooling system achieved for the considered configurations.

Comparing the referring case to the cooled ones, the cooling effect is considerable; moreover the overall heat losses coefficient $U_{\text{pan-system}}$ enhances from the first to the fourth case accordingly with the reduction of the mean cell temperature goes from 320.35 K for the reference case to 308.25 K for the Case A4. Fig. 9 shows different cell temperature profiles along the center of the PV panel for each considered PV – air cooling system solutions.

The minimum temperature value in each case corresponds to the centreline of the duct partition; indeed it represents a preferential way for the heat. In regards to the cases with three ducts, it is important to notice a moderate decrease of cell temperature values profile incrementing the sheet thickness (Fig. 9), although the value of $U_{\text{pan-system}}$ is almost the same (Table 5). Moreover, the metal sheet decreases the maximum temperature value more than

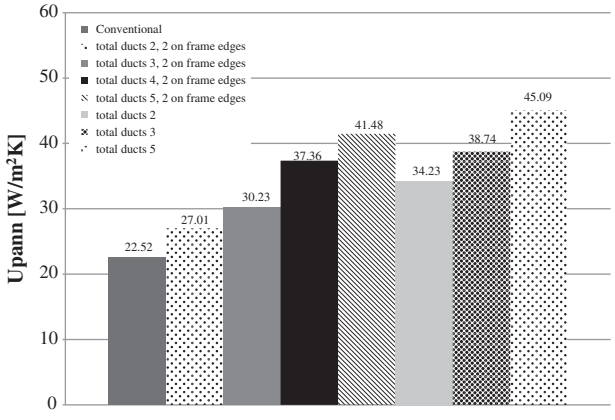


Fig. 8. PV-cooling water system overall heat losses coefficient of the considered solution concerning the optimization phase.

Table 5
Air cooling system configurations. Mean cell temperature T_{cell}, overall heat loss coefficient U_{pan} and thermal capacity C_t of the different configurations of cooled PV panels.

	T_{cell} (K)	$U_{\text{pan-system}}$ (W/(m^2 K))	C_t (J/(m^2 K))
Reference case	320.35	22.52	10,184
Case A1	315.15	27.90	12,614
Case A2	311.95	32.56	15,084
Case A3	311.35	33.65	17,474
Case A4	308.25	40.60	15,084

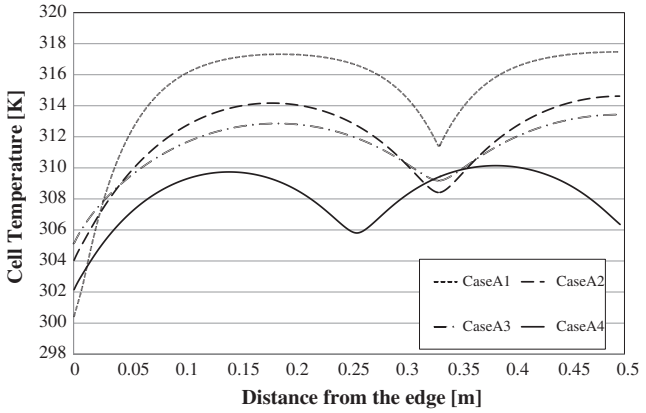


Fig. 9. Temperature profiles of the PV cell with different air cooling system configuration.

3 K. To avoid an extreme increment of the thermal capacity of the whole system, a thickness of 1 mm of metal sheet has been chosen for the fourth case with 4 ducts. A temperature difference of 4 K, from the edge to the center of the partition, has been achieved in the A4 case, therefore this last configuration has been considered good enough for the purpose of the research, reaching a close temperature difference achieved with the water cooling system.

4.3. Air speed field profile

The comparison between measured and simulated air speed in terms of mean values, measured punctual speeds and simulated speed profiles is shown in Fig. 10.

The reported simulated mean speed regards only the profile through the measured points (from 0.035 m to 0.215 m). The points in Fig. 10 concern the measured speeds in the related sections (Fig. 4). The best accordance has been recorded in Section 2, in which the speed is less affected by the inlet and outlet conditions. Indeed, the

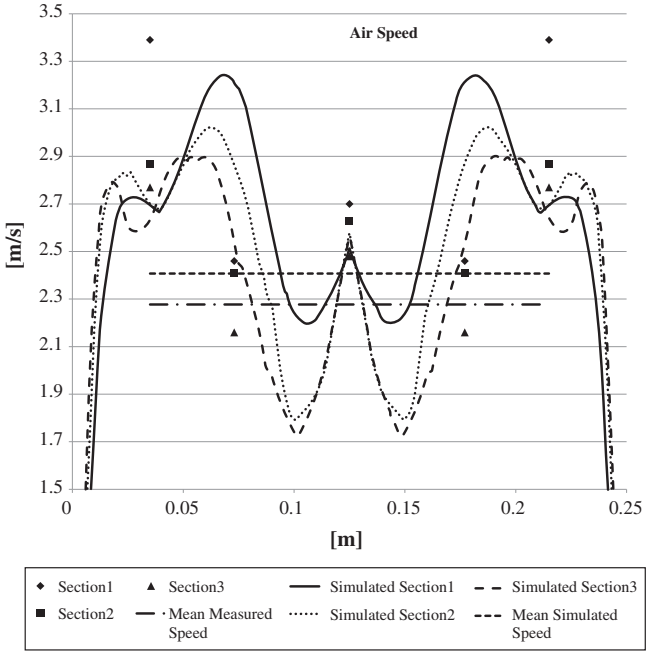


Fig. 10. Air cooling system, comparison between the simulated profiles and the measured values of air speed field in three different sections of the PV panel.

effects produced by the fan, which have not been simulated, are far away from the constant speed condition used in the simulation. However in the energy analysis, goal of this investigation, the correlations used in the finite element software to estimate the convention coefficient of the cooling system depend upon the mean speed of the working fluid. Hence, as Fig. 10 shows, the real speed profile is well described by the simulation and, moreover, the mean speed presents an estimated error of 5.4% confirming that the related convective coefficient is realistic.

4.4. Energy analysis

The energy analysis resulted have been conducted only for the identified best PV – cooling system cases having as a working fluid air and water:

- PV-water cooling system with 5 cooling ducts
- PV-air cooling system with 4 cooling ducts, Case A4.

To show the benefits of the best identified systems the results have been compared with the conventional PV panel and with the water system Case 9′ performances as well, which has the best absolute performance, but an extreme weight. Fig. 11 shows the efficiency trends of the panel along two typical clear and cloudy days of summer and winter conditions for Cosenza. The highest benefits have been recorded in the clear winter and summer days, in which high cells temperatures without cooling system considerably decrease the PV efficiency.

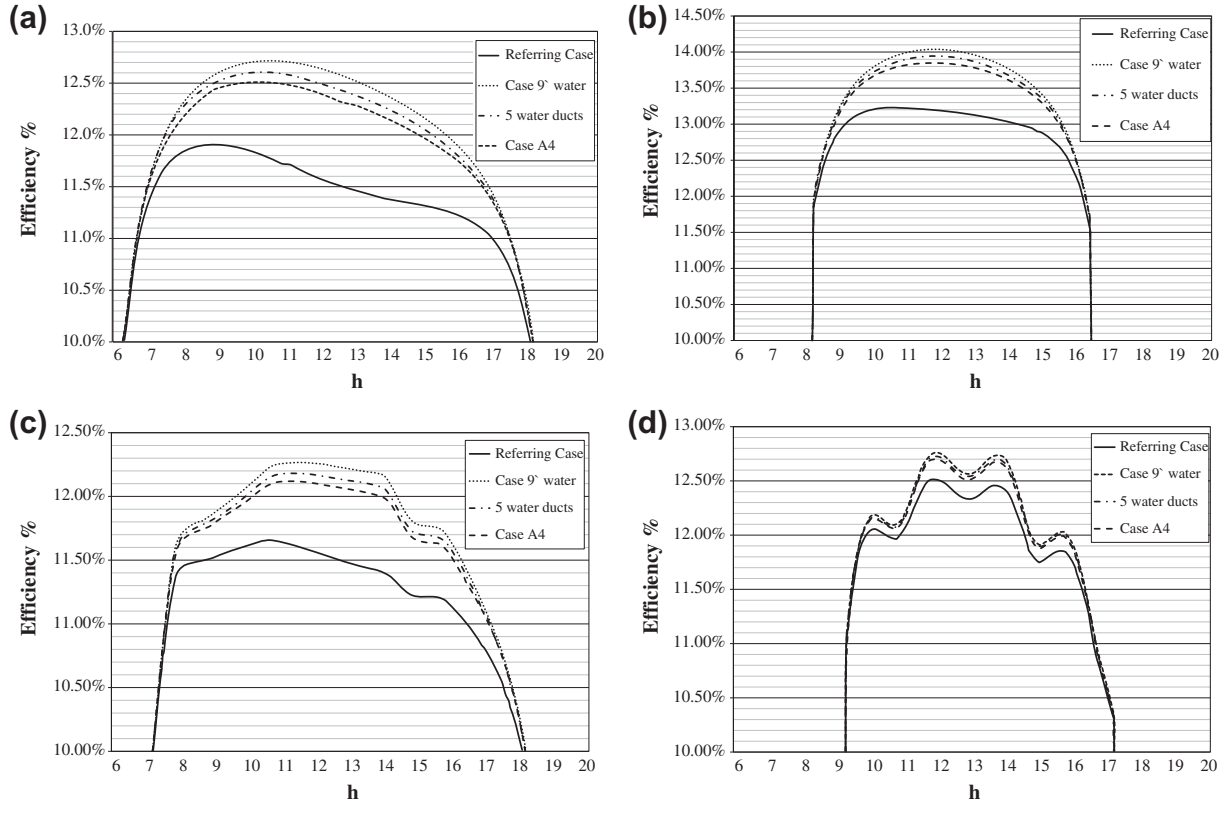


Fig. 11. Efficiency trend of the PV cells for the climatic conditions of Cosenza. Four cases have been considered: the reference one without cooling system and the best identified configurations with water and air cooling systems. (a) Clear summer day, (b) clear winter day, (c) cloudy summer day, and (d) cloudy winter day.

Table 6
Energy analysis results: monthly and yearly energy produced, percentage energy gained from the reference case and yearly mean efficiency of the reference case, 5 water ducts and Case 4A PV-cooling systems.

	Energy (Wh)				
	Reference case	5 Water ducts		Case A4	
Jan	12,663	13,138	3.76%	13,087	3.35%
Feb	16,268	16,968	4.30%	16,892	3.83%
Mar	23,208	24,316	4.77%	24,195	4.26%
Apr	24,580	25,808	5.00%	25,675	4.45%
May	26,891	28,271	5.13%	28,121	4.58%
Jun	28,175	29,748	5.58%	29,577	4.98%
Jul	28,468	30,086	5.68%	29,910	5.06%
Aug	27,870	29,474	5.75%	29,300	5.13%
Sep	23,549	24,811	5.36%	24,674	4.78%
Oct	17,940	18,761	4.58%	18,672	4.08%
Nov	14,793	15,410	4.17%	15,343	3.71%
Dec	13,612	14,138	3.87%	14,081	3.45%
TOT (kW h)	258.02	270.93		269.53	
Mean yearly efficiency (%)	12.04	12.65		12.58	

At noon of the clear summer day 0.82% efficiency points have been gained by using Case 4A cooling system configuration compared to the reference case, instead 0.92% points using 5 cooling water duct configuration (Fig. 11a). Instead concerning the clear winter day 0.65% and 0.73% points have been gained at noon by means of Case 4A and cooling water duct configurations winter clear days (Fig. 11b). Concerning the cloudy summer day, 0.54% and 0.61% efficiency points have been gained using Case 4A and cooling water duct configurations, showing that the cooling system is suitable in summer cloudy days as well. Instead few benefits have been recorded in the winter cloudy day (Fig. 11d), particularly only 0.19% and 0.22% efficiency points has been gained using Case 4A and cooling water duct configurations.

The monthly produced energy of the reference case, the 5 water ducts and the Case A4 PV cooling-systems are reported in Table 6, which shows the monthly percentage increments between the reference case and the related PV-cooling system, the yearly energy produced and the yearly mean efficiency. The monthly energy results state the same conclusion of the efficiency analysis (Fig. 11), indeed both systems, 5 water ducts and Case A4, gain more in summer than in winter.

Both 5 water ducts and Case A4 PV – cooling systems mean yearly efficiency and yearly produced energy are almost the same, respectively 12.65%, 12.58% and 270.93 kW h, 269.53 kW h (Table 6).

However, to improve the winter performances the air analyzed cooling system could be potentially used in BIPV Thermal (BIPVT) applications to decrease the space heating energy requirements using the extracted heat and improving, at the same time, the whole PV–thermal system efficiency (Dubey et al., 2009; Agrawal and Tiwari, 2010; Norton et al., 2011; Kamthania et al., 2011). Instead, in cooling dominated climate, with high external air temperature and irradiance level, the system shows better performance and it might be useful to increase the production. Moreover regarding BIPVT application, it can also reduce cooling building requirements blocking the heat gain through the building envelope if used as a ventilated wall (Yang et al., 2000; Chow et al., 2007; Peng et al., 2013).

Further investigations aiming to find the best configuration for this particular system for BIPVT application are needed.

5. Conclusions

In this study different PV-cooling systems with water and air have been studied. The water system is more complex than the air system; indeed it needs a watertight pumping device, while the air system is assembled with simple fans. Different construction aspects have been analyzed: use of a metal sheet between the PV module and the cooling system, different cooling ducts materials and ducts pitches for the water system and different number of ducts for the air system.

In regards to the water system, the analysis have demonstrated that the use of plastic material for the cooling ducts, despite they present significant construction advantages, offers low thermal performance, therefore metallic ducts with higher thermal conductibility are requested. The number of ducts can be reduced using a metal sheet maintaining a low temperature difference between the ducts, maximum variation of 3 K has been achieved with a duct pitch of 159 mm (Fig. 7) corresponding to an overall heat loss coefficient of the PV – water system of 45.09 W/m^2 K (Fig. 8).

Concerning the air system the use of the metal sheet produces an almost flat cell temperature profile. The air speed field has been verified comparing the simulated results with the measured profile demonstrating that it is possible to achieve the considered speed using a simple fan for each duct. In this case the maximum temperature variation achieved with 4 ducts (Fig. 3) is 4 K corresponding to an overall heat loss coefficient of the PV – water system of 40.60 W/m^2 K (Table 5).

Only the best identified air and water cooling solutions, 5 water ducts and Case A4 PV – cooling systems, have been considered in the energy analyses carried out by TRNSYS simulation code. Simulations results show similar performances of both water and air systems, respectively a mean yearly efficiency of 12.65%, 12.58% and a produced energy of 270.93 kW h, 269.53 kW h (Table 6). Both show better result, in terms of energy production and gained percentage efficiency points, in summer compared to the reference case (Table 6). The yearly improvement of a conventional PV system with the considered air or water cooling system is about 5%, further improvement can be achieved especially regarding the air system if the system is used in BIPV applications.

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References

Agrawal, Basant, Tiwari, G.N., 2010. Life cycle cost assessment of building integrated photovoltaic thermal (BIPVT) systems. Energy Build. 42, 1472–1481.

Amery, M., Abdolizadeh, M., 2009. Improving the effectiveness of a photovoltaic water pumping system by spraying water over the front of photovoltaic cells. Renewable Energy 34, 91–96.

Anderson, T.N., Duke, M., Morrison, G.L., Carson, J.K., 2009. Performance of a building integrated photovoltaic/thermal (BIPVT) solar collector. Solar Energy 83, 445–455.

Assoa, Y.B., Menezo, C., Fraisse, G., Yezou, R., Brau, J., 2007. Study of a new concept of photovoltaic–thermal hybrid collector. Solar Energy 81, 1132–1143.

Bambrook, S.M., Sproul, A.B., 2012. Maximising the energy output of a PVT air system. Solar Energy 86, 1857–1871.

Bejan, A., 1993. Heat Transfer. Wiley.

Brinkworth, B.J., 2006. Optimum depth for PV cooling ducts. Solar Energy 80, 1131–1134.

Brinkworth, B.J., Cross, B.M., Marshall, R.H., Yang, H., 1997. Thermal regulation of photovoltaic cladding. Solar Energy 61, 169–178.

Chow, T.T., He, W., Ji, J., 2007. An experimental study of façade-integrated photovoltaic/water-heating system. Appl. Therm. Eng. 27, 37–45.

Cucumo, M.A., Marinelli, V., Oliveti, G., 1994. Ingegneria Solare principi e applicazioni, Pitagora Editrice Bologna.

Dubey, Swapnil, Solanki, S.C., Tiwari, Arvind, 2009. Energy and exergy analysis of PV/T air collectors connected in series. Energy Build. 41, 863–870.

Ghani, F., Duke, M., Carson, J.K., 2012a. Effect of flow distribution on the photovoltaic performance of a building integrated photovoltaic/thermal (BIPV/T) collector. Solar Energy 86, 1518–1530.

Ghani, F., Duke, M., Carson, J.K., 2012b. Estimation of photovoltaic conversion efficiency of a building integrated photovoltaic/thermal (BIPV/T) collector array using an artificial neural network. Solar Energy 86, 3378–3387.

Hiyama, T., Kouzuma, S., Imakudo, T., 1995. Identification of optimal operating point of PV modules using neural networks for real time maximum power point tracking control. IEEE Trans. Energy Convers. 10, 360–367.

Hua, C., Lin, J., Shen, C., 1998. Implementation of a DSP-controlled photovoltaic system with peak power tracking. IEEE Trans. Energy Convers. 45, 99–107.

Incropera, F.P., DeWitt, D.P., 1996. Fundamentals of Heat and Mass Transfer, fourth ed. John Wiley & Sons.

Joshi, J.C., 1994. Thermal evaluation of different types of solar cell modules in sunlight. Renewable Energy 5, 1761–1763.

Kamthania, Deepali, Nayak, Sujata, Tiwari, G.N., 2011. Performance evaluation of a hybrid photovoltaic thermal double pass facade for space heating. Energy Build. 43, 2274–2281.

Krauter, S., 2004. Increased electrical yield via water flow over the front of photovoltaic panels. Solar Energy Mater. Solar Cells 82, 131–137.

Malik, A.Q., Bin Haji Damit, Salmi Jan, 2003. Outdoor testing of single crystal silicon solar cells. Renewable Energy 28, 1433–1445.

Marinelli, V., Oliveti, G., Sabato, A., 1984. Elementi di Trasmissione del calore, Pitagora Editrice Bologna.

TRNSYS Mathematical Reference, Solar Energy Laboratory University of Wisconsin–Madison, 2006. vol. 5. pp. 85–89.

Mishra, R.K., Tiwari, G.N., 2013. Energy matrices analyses of hybrid photovoltaic thermal (HPVT) water collector with different PV technology. Solar Energy 91, 161–173.

Norton, Brian, Eames, Philip C., Mallick, Tapas K., Huang, Ming Jun, McCormack, Sarah J., Mondol, Jayanta D., Yohanis, Yigzaw G., 2011. Enhancing the performance of building integrated photovoltaics. Solar Energy 85, 1629–1664.

Peng, Jinqing, Lin, Lu, Yang, Hongxing, Han, Jun, 2013. Investigation on the annual thermal performance of a photovoltaic wall mounted on a multi-layer façade. Appl. Energy 112, 646–656.

Probes Air speed Manual, SI® Company, 2005.

Raziemska, H.E., 2003. The effect of temperature on the power drop in crystalline silicon solar cells. Renewable Energy 28, 1–12.

Skoplaki, E., Palyvos, J.A., 2009. On the temperature dependence of photovoltaic module electrical performance: a review of efficiency/power correlations. Solar Energy 83, 614–624.

TRNSYS, Version 16, Solar Energy Laboratory University of Wisconsin–Madison, 2006.

Yang, Hongxing, Burnett, John, Ji, Jie, 2000. Simple approach to cooling load component calculation through PV walls. Energy Build. 31, 285–290.

Ye, Zhen, Nobre, André, Reindl, Thomas, Luther, Joachim, Reise, Christian, 2013. On PV module temperatures in tropical regions. Solar Energy 88, 80–87.

Yin, H.M., Yang, D.J., Kelly, G., Garant, J., 2013. Design and performance of a novel building integrated PV/thermal system for energy efficiency of buildings. Solar Energy 87, 184–195.

Yun, Geun Young, McEvoy, Mike, Steemers, Koen, 2007. Design and overall energy performance of a ventilated photovoltaic façade. Solar Energy 81, 383–394.